نظام إدارة درجات الطلاب

Student Grades Management System — Topic 3: Programming (Python)

نتائج التشغيل والاختبارات – Run & Test Results

جميع المخرجات في هذا الملف منسوخة كما هي من تشغيل فعلي للبرنامج على Python 3.13، وليست مكتوبة يدويًا. وبجانب كل سؤال في المخرجات ما أُدخل.

1. ملخص حالات الاختبار

#	الحالة	المدخلات	المتوقع	الفعلي	الحكم
1	طالب ناجح	Ahmed Ali: 85, 90, 78	253 / 84.33 / ناجح	**PASS** / 253.0 / 84.33	صحيح ✓
2	طالب راسب	Sara Hassan: 40, 35, 50	125 / 41.67 / راسب	**FAIL** / 125.0 / 41.67	صحيح ✓
3	حد النجاح بالضبط	Khalid: 50, 50, 50	50.00 / ناجح	**PASS** / 50.00	صحيح ✓
4	أقل من الحد بقليل	Mona: 49, 50, 50	49.67 / راسب	**FAIL** / 49.67	صحيح ✓
5	مدخلات خاطئة	اسم فارغ، Omar99، abc، 120، -5، maybe	رسالة خطأ لكل إدخال خاطئ	خطآن في الاسم، 3 في الدرجة، 1 في الإجابة، ثم **PASS**	صحيح ✓
6	خمسة طلاب	انظر التشغيل 6	3 ناجحون، 2 راسبان، متوسط 59.20	3 / 2 / 59.20	صحيح ✓

2. مخرجات التشغيل الفعلي

التشغيل 1: طالب ناجح

```
================================================================================
                        STUDENT GRADES MANAGEMENT SYSTEM
         Subjects: Math, Science, English  |  Pass mark (average): 50
================================================================================

--- Student #1 ---
Enter student name: Ahmed Ali
  Enter Math grade (0-100): 85
  Enter Science grade (0-100): 90
  Enter English grade (0-100): 78
  -> Total = 253.0, Average = 84.33, Status = PASS
Add another student? (y/n): n

================================================================================
                                STUDENT RESULTS
================================================================================
No. Name                    Math  Science  English    Total  Average  Status
--------------------------------------------------------------------------------
1   Ahmed Ali               85.0     90.0     78.0    253.0    84.33    PASS
--------------------------------------------------------------------------------
Number of students : 1
Passed             : 1
Failed             : 0
Class average      : 84.33
================================================================================
Program finished. Goodbye!
```

التشغيل 2: طالب راسب

```
================================================================================
                        STUDENT GRADES MANAGEMENT SYSTEM
         Subjects: Math, Science, English  |  Pass mark (average): 50
================================================================================

--- Student #1 ---
Enter student name: Sara Hassan
  Enter Math grade (0-100): 40
  Enter Science grade (0-100): 35
  Enter English grade (0-100): 50
  -> Total = 125.0, Average = 41.67, Status = FAIL
Add another student? (y/n): n

================================================================================
                                STUDENT RESULTS
================================================================================
No. Name                    Math  Science  English    Total  Average  Status
--------------------------------------------------------------------------------
1   Sara Hassan             40.0     35.0     50.0    125.0    41.67    FAIL
--------------------------------------------------------------------------------
Number of students : 1
Passed             : 0
Failed             : 1
Class average      : 41.67
================================================================================
Program finished. Goodbye!
```

التشغيل 3: حد النجاح بالضبط (المتوسط 50)

```
================================================================================
                        STUDENT GRADES MANAGEMENT SYSTEM
         Subjects: Math, Science, English  |  Pass mark (average): 50
================================================================================

--- Student #1 ---
Enter student name: Khalid
  Enter Math grade (0-100): 50
  Enter Science grade (0-100): 50
  Enter English grade (0-100): 50
  -> Total = 150.0, Average = 50.00, Status = PASS
Add another student? (y/n): n

================================================================================
                                STUDENT RESULTS
================================================================================
No. Name                    Math  Science  English    Total  Average  Status
--------------------------------------------------------------------------------
1   Khalid                  50.0     50.0     50.0    150.0    50.00    PASS
--------------------------------------------------------------------------------
Number of students : 1
Passed             : 1
Failed             : 0
Class average      : 50.00
================================================================================
Program finished. Goodbye!
```

التشغيل 4: أقل من الحد بقليل (المتوسط 49.67)

```
================================================================================
                        STUDENT GRADES MANAGEMENT SYSTEM
         Subjects: Math, Science, English  |  Pass mark (average): 50
================================================================================

--- Student #1 ---
Enter student name: Mona
  Enter Math grade (0-100): 49
  Enter Science grade (0-100): 50
  Enter English grade (0-100): 50
  -> Total = 149.0, Average = 49.67, Status = FAIL
Add another student? (y/n): n

================================================================================
                                STUDENT RESULTS
================================================================================
No. Name                    Math  Science  English    Total  Average  Status
--------------------------------------------------------------------------------
1   Mona                    49.0     50.0     50.0    149.0    49.67    FAIL
--------------------------------------------------------------------------------
Number of students : 1
Passed             : 0
Failed             : 1
Class average      : 49.67
================================================================================
Program finished. Goodbye!
```

التشغيل 5: مدخلات خاطئة والتحقق منها

```
================================================================================
                       STUDENT GRADES MANAGEMENT SYSTEM
         Subjects: Math, Science, English  |  Pass mark (average): 50
================================================================================

--- Student #1 ---
Enter student name:
  [Error] Name must not be empty and must contain letters only.
Enter student name: Omar99
  [Error] Name must not be empty and must contain letters only.
Enter student name: Omar Saleh
  Enter Math grade (0-100): abc
  [Error] Grade must be a number between 0 and 100.
  Enter Math grade (0-100): 120
  [Error] Grade must be a number between 0 and 100.
  Enter Math grade (0-100): -5
  [Error] Grade must be a number between 0 and 100.
  Enter Math grade (0-100): 60
  Enter Science grade (0-100): 70.5
  Enter English grade (0-100): 80
  -> Total = 210.5, Average = 70.17, Status = PASS
Add another student? (y/n): maybe
  [Error] Please enter 'y' or 'n'.
Add another student? (y/n): n

================================================================================
                              STUDENT RESULTS
================================================================================
No. Name                    Math  Science  English    Total  Average  Status
--------------------------------------------------------------------------------
1   Omar Saleh              60.0     70.5     80.0    210.5    70.17    PASS
--------------------------------------------------------------------------------
Number of students : 1
Passed             : 1
Failed             : 0
Class average      : 70.17
================================================================================
Program finished. Goodbye!
```

التشغيل 6: خمسة طلاب (فصل كامل)

```
================================================================================
                        STUDENT GRADES MANAGEMENT SYSTEM
         Subjects: Math, Science, English  |  Pass mark (average): 50
================================================================================

--- Student #1 ---
Enter student name: Ahmed Ali
  Enter Math grade (0-100): 85
  Enter Science grade (0-100): 90
  Enter English grade (0-100): 78
  -> Total = 253.0, Average = 84.33, Status = PASS
Add another student? (y/n): y

--- Student #2 ---
Enter student name: Sara Hassan
  Enter Math grade (0-100): 40
  Enter Science grade (0-100): 35
  Enter English grade (0-100): 50
  -> Total = 125.0, Average = 41.67, Status = FAIL
Add another student? (y/n): y

--- Student #3 ---
Enter student name: Khalid
  Enter Math grade (0-100): 100
  Enter Science grade (0-100): 100
  Enter English grade (0-100): 100
  -> Total = 300.0, Average = 100.00, Status = PASS
Add another student? (y/n): Y

--- Student #4 ---
Enter student name: Mona
  Enter Math grade (0-100): 0
  Enter Science grade (0-100): 20
  Enter English grade (0-100): 30
  -> Total = 50.0, Average = 16.67, Status = FAIL
Add another student? (y/n): y

--- Student #5 ---
Enter student name: Fatimah
  Enter Math grade (0-100): 55
  Enter Science grade (0-100): 45
  Enter English grade (0-100): 60
  -> Total = 160.0, Average = 53.33, Status = PASS
Add another student? (y/n): no

================================================================================
                                STUDENT RESULTS
================================================================================
No. Name                   Math  Science  English    Total  Average  Status
--------------------------------------------------------------------------------
1   Ahmed Ali              85.0     90.0     78.0    253.0    84.33    PASS
2   Sara Hassan            40.0     35.0     50.0    125.0    41.67    FAIL
3   Khalid                100.0    100.0    100.0    300.0   100.00    PASS
4   Mona                    0.0     20.0     30.0     50.0    16.67    FAIL
5   Fatimah                55.0     45.0     60.0    160.0    53.33    PASS
--------------------------------------------------------------------------------
Number of students : 5
Passed             : 3
Failed             : 2
Class average      : 59.20
```

```
================================================================================
Program finished. Goodbye!
```

3. اختبارات الوحدة الآلية

الأمر: `python3 -m unittest -v test_grades_system` — نجحت 11 من 11.

```
test_invalid_inputs_are_rejected_then_accepted
(test_grades_system.TestFullProgram.test_invalid_inputs_are_rejected_then_accepted) ... ok
test_pass_and_fail_students (test_grades_system.TestFullProgram.test_pass_and_fail_students) ...
ok
test_record (test_grades_system.TestProcessing.test_record) ... ok
test_status_boundary (test_grades_system.TestProcessing.test_status_boundary) ... ok
test_status_fail (test_grades_system.TestProcessing.test_status_fail) ... ok
test_status_pass (test_grades_system.TestProcessing.test_status_pass) ... ok
test_total_and_average (test_grades_system.TestProcessing.test_total_and_average) ... ok
test_invalid_grades (test_grades_system.TestValidation.test_invalid_grades) ... ok
test_invalid_names (test_grades_system.TestValidation.test_invalid_names) ... ok
test_valid_grades (test_grades_system.TestValidation.test_valid_grades) ... ok
test_valid_names (test_grades_system.TestValidation.test_valid_names) ... ok

----------------------------------------------------------------------
Ran 11 tests in 0.002s

OK
```

4. الخلاصة

- البرنامج يعمل دون أخطاء، ويحسب المجموع والمتوسط بدقة.
- تحديد الحالة صحيح، بما في ذلك الحالات الحدّية (50 ناجح، 49.67 راسب).
- كل المدخلات الخاطئة تُرفض برسالة واضحة ويُعاد السؤال دون توقف البرنامج.
- يمكن إدخال أي عدد من الطلاب، وتُعرض النتائج في جدول منظّم مع ملخص الفصل.